

A-level

# ENVIRONMENTAL SCIENCE

## Questions by topic: Manipulation of food species & agricultural energetics - Set 1

### Mark scheme

AQA spec 3.5.1.2; Biological resources; textbook pp. 321-327

### Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

**Textbook references:** every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Marks |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>01.1</b> | <ul style="list-style-type: none"> <li>• Selective breeding: parents chosen for desirable genetic characteristics (Limousin muscle growth, Merino wool, Cheviot hardiness); combines genes but characteristics of offspring not fully predictable; many generations needed; risk of inbreeding</li> <li>• Asexual reproduction (cuttings, runners, tissue culture, cloning): offspring genetically identical to the parent so characteristics are predictable and survival high; but no variation so no improvement and vulnerability to the same disease; cloning of animals (nucleus transfer into an enucleated egg, surrogate) still developing</li> <li>• Cross-breeding: two breeds combined to give a mix of characteristics with hybrid vigour (heterosis) and lower inbreeding risk, eg Zebu (heat tolerant) x Ayrshire (high milk yield)</li> </ul> | 6     |
| <b>01.2</b> | <ul style="list-style-type: none"> <li>• AI: semen from a male with desirable characteristics can fertilise many more females than natural mating; frozen semen stored and transported easily and used after the male dies</li> <li>• Embryo transfer: FSH stimulates a superior female to release many eggs, which are fertilised (IVF) with sperm from a superior male</li> <li>• Each embryo is implanted in a different surrogate female</li> <li>• So the donor produces many offspring per year rather than one per pregnancy</li> </ul>                                                                                                                                                                                                                                                                                                                | 4     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-323; AQA spec 3.5.1.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |       |
| <b>02.1</b> | <ul style="list-style-type: none"> <li>• Single genes can be transferred from any species, not just closely related ones that produce fertile hybrids</li> <li>• Desired characteristics introduced without linked unwanted ones; faster</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2     |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 02.2     | <ul style="list-style-type: none"> <li>• Bt: reduces insecticide spraying so fewer non-target deaths, lower cost / energy</li> <li>• Roundup Ready: weeds controlled by spraying glyphosate without harming the crop, so higher yields with less cultivation / tillage</li> <li>• Golden rice: provides vitamin A in the staple diet, reducing about 500 000 cases of blindness a year in India / sub-Saharan Africa</li> </ul>                                                                                                                                                                                                  | 3     |
| 02.3     | <ul style="list-style-type: none"> <li>• Gene transfer by pollen to conventional / organic crops or wild relatives (eg Bt gene killing insects feeding on wild plants)</li> <li>• Antibiotic-resistance marker genes could transfer to pathogenic bacteria; possible allergies; gene transfer to gut bacteria</li> <li>• Detection in the human food chain from crops approved only for animal feed</li> <li>• Economic: seeds must be bought each year from patent-holding companies, raising costs and reducing local crop diversity and seed saving in LEDCs; herbicide-resistant crops may increase herbicide use</li> </ul> | 4     |
| 02.4     | <ul style="list-style-type: none"> <li>• Public / political concern that risks are not fully assessed - more evidence needed (precautionary principle)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 1     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 323-324; AQA spec 3.5.1.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |       |
| 03.1     | <ul style="list-style-type: none"> <li>• Extensive cereal 5.0; intensive cereal 3.0</li> <li>• Intensive beef 0.1</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2     |
| 03.2     | <ul style="list-style-type: none"> <li>• Intensive cereal is most productive per hectare (60 GJ) but less efficient (3.0) than extensive (5.0)</li> <li>• Each extra unit of input gives a smaller extra yield (law of diminishing returns)</li> <li>• Beef has very low productivity and efficiency because energy is lost at an extra trophic level</li> </ul>                                                                                                                                                                                                                                                                 | 3     |
| 03.3     | <ul style="list-style-type: none"> <li>• Any input that increases productivity but requires the use of energy (usually fossil fuel)</li> <li>• Manufacture of nitrate fertiliser (Haber process) and pesticides; fuel for machinery; pumping irrigation water; grain drying; food processing and transport; embodied energy of machinery</li> </ul>                                                                                                                                                                                                                                                                              | 3     |
| 03.4     | <ul style="list-style-type: none"> <li>• Because of diminishing returns, the first units of input on unfertilised land give the largest yield increases</li> <li>• Moving inputs from intensively farmed land (where extra input adds little) to extensively farmed land gives a greater total yield - though land, cost and distribution limit this</li> </ul>                                                                                                                                                                                                                                                                  | 2     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 324-327; AQA spec 3.5.1.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |       |
| 04.1     | <ul style="list-style-type: none"> <li>• Chicken and salmon need far less feed per kg gained (1.2-1.8 vs 7.0)</li> <li>• Fish are ectotherms so no energy is used for thermoregulation and water supports their mass; poultry grow fast and are kept warm indoors with movement restricted</li> <li>• Cattle are large endotherms, use energy in movement, and lose energy in methane from cellulose digestion</li> </ul>                                                                                                                                                                                                        | 3     |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Marks |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 04.2     | <ul style="list-style-type: none"> <li>Control of movement (housing / restricted grazing) so less energy is used in muscle activity</li> <li>Temperature control (heated / insulated buildings) so less energy is used in maintaining body temperature</li> <li>Species selection: choose species with low FCRs (poultry, fish); high-energy easily digested diets; slaughter at the fastest growth stage</li> </ul>                                                                                                                                                                                                                                            | 3     |
| 04.3     | <ul style="list-style-type: none"> <li>Only about 10% of the energy at one trophic level is passed to the next; most is lost as heat in respiration / metabolism or in undigested material</li> <li>Humans eating plants (primary consumers) receive more of the energy captured by photosynthesis than eating animals fed on those plants (secondary consumers)</li> <li>Livestock better where land cannot grow crops (upland, semi-arid, steep, thin soils): grazing animals with symbiotic gut bacteria digest cellulose in grass that humans cannot</li> <li>Omnivores (pigs) convert food wastes (dairy by-products, catering waste) into meat</li> </ul> | 4     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 326-328; AQA spec 3.5.1.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |
| 05.1     | <ul style="list-style-type: none"> <li>Cross-bred offspring are healthier / more productive because the problems caused by homozygous recessive genes in inbred varieties are reduced (heterosis)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1     |
| 05.2     | <ul style="list-style-type: none"> <li>Zebu tolerate heat but have a low milk yield; Ayrshire have a high milk yield but are not heat tolerant</li> <li>Cross-breeding combined the desirable characteristics: heat tolerance with a high milk yield</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                 | 2     |
| 05.3     | <p><i>Any 2 from:</i></p> <ul style="list-style-type: none"> <li>Replacement of valuable animals that die with genetically identical individuals</li> <li>Replacement of herds culled during a disease outbreak</li> <li>Production of large numbers of individuals with desirable characteristics</li> </ul>                                                                                                                                                                                                                                                                                                                                                   | 2     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-322; AQA spec 3.5.1.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |
| 06.1     | <ul style="list-style-type: none"> <li><math>1000 / 100\,000 \times 100 = 1\%</math></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1     |
| 06.2     | <ul style="list-style-type: none"> <li>Some light is reflected by the leaves</li> <li>Some light is transmitted (passes through the leaves without being absorbed) or is of wavelengths not absorbed</li> <li>Much of the energy captured is used in the plant's metabolic processes and released as low energy-density heat (respiration)</li> </ul>                                                                                                                                                                                                                                                                                                           | 3     |
| 06.3     | <ul style="list-style-type: none"> <li>Less energy is passed to each trophic level than was received from the previous one (heat losses in respiration)</li> <li>After four levels too little energy remains to support a further population</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                         | 2     |
| 06.4     | <ul style="list-style-type: none"> <li>Permanent grassland is a more sustainable land use than arable crops in these areas (soil / rainfall unsuitable for crops)</li> <li>Grazing animals can digest grass cellulose using symbiotic bacteria in their stomachs, producing food that humans can eat</li> </ul>                                                                                                                                                                                                                                                                                                                                                 | 2     |
| 06.5     | <ul style="list-style-type: none"> <li>Pigs are omnivores that can eat food wastes not wanted by people (eg cheese-making wastes, catering leftovers)</li> <li>So edible meat is produced from material that would otherwise be wasted, without using land / crops that could feed people</li> </ul>                                                                                                                                                                                                                                                                                                                                                            | 2     |

| Question    | Marking guidance                                                                                                                                                                                                                                                                           | Marks |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 326-328; AQA spec 3.5.1.2.                                                                                                                                                                    |       |
| <b>07.1</b> | <ul style="list-style-type: none"> <li>• Yield increases as inputs increase</li> <li>• But each extra unit of input produces a smaller increase in yield than the previous one (the curve flattens / law of diminishing returns)</li> </ul>                                                | 2     |
| <b>07.2</b> | <ul style="list-style-type: none"> <li>• Second unit: <math>70 - 50 = 20</math> units</li> <li>• Fifth unit: <math>94 - 90 = 4</math> units</li> </ul>                                                                                                                                     | 2     |
| <b>07.3</b> | <ul style="list-style-type: none"> <li>• The first units of input on each area give the largest yield increases</li> <li>• Concentrating inputs means later units add very little, whereas the same inputs spread thinly all work at the steep part of the curve</li> </ul>                | 2     |
| <b>07.4</b> | <ul style="list-style-type: none"> <li>• Some areas have much greater potential productivity so investment there is sensible</li> <li>• Affluent societies can afford to buy more inputs while poorer communities that would benefit most cannot afford them (economic reasons)</li> </ul> | 2     |
| <b>07.5</b> | <ul style="list-style-type: none"> <li>• Yield per unit area is very high because of large inputs</li> <li>• But the extra yield per unit of input is small, so efficiency (yield per unit input / energy ratio) is low</li> </ul>                                                         | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 324-325; AQA spec 3.5.1.2.                                                                                                                                                                    |       |